<LSDS PROJECT>

PROJECT RISK MANAGEMENT PLAN

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<06/02/2025>

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VERSION HISTORY

Version #	Implemented By	Revision Date	Approved By	Approval Date	Reason
1.0	Sibin Jacob	4/6/25	<Sibin Jacob>	4/6/25	Sections 1, 2, 3, and 9
2.0	Sibin Jacob	4/20/25	Sibin Jacob	4/22/25	Complete sections 6, 7, 8
3.0	Sibin Jacob	4/27/25	Sibin Jacob	4/27/25	Updated columns 6,7, & 8 of the risk register. Completed Sections 4 & 5.
4.0	Sibin Jacob	5/10/25	Sibin Jacob	5/10/25	Completed columns 9 & 10 of the Risk Register. Updated Appendix B and adding explanations on risk attitude columns under "Stakeholder Risk Tolerance" table.
5.0	Sibin Jacob	5/18/25	Sibin Jacob	5/18/25	Added information to "Tracking" and "Reporting Formats" sections.
6.0	Sibin Jacob	6/2/25	Sibin Jacob	6/2/25	Signed for approval. Completed Quality Methodologies section.
7.0	Sibin Jacob	6/2/25	Sibin Jacob	6/2/25	Final signatures. Updated risk status, budget, and time contingency tables.

TABLE OF CONTENTS

INTRODUCTION

LSDS contracted with ASDG to develop and deliver a project risk management training to train team members and produce the following outputs and activities:

1. Develop a LSDS project management plan
2. Document a risk register, tracks and prioritizes risks or uncertainties, either resulting in positive or negative results.
3. Perform Qualitative and Quantitative Risk Analysis
4. Risk management to improve quality
5. Monitor, control, and manage risk events when they become issues

The overall objective of this Project Risk Management Plan is to help ensure a common understanding among project stakeholders of how uncertainties for the project are planned to be identified, evaluated, categorized, prioritized, documented, and managed for the *LSDS* project.

This Project Risk Management Plan defines the roles and responsibilities to be undertaken, explains the processes to be used, identifies uncertainties, and provides a plan for managing uncertainties associated with the client's project. It essentially outlines an understanding to be achieved among stakeholders regarding the processes and actions necessary to facilitate ensuring uncertainties do not preclude project success.

The intended audience of the *LSDS* Project Risk Management Plan is the client project sponsor, contractor program manager, project manager, project team, senior leaders, and any other stakeholder whose support is needed to carry out the planning and managing of uncertainties for the client's project.

PROJECT RISK MANAGEMENT PROCESSES & METHODOLOGY

Risk Management Process/Activity	Risk Management Process / Activity Description	Ownership
Risk Identification	Risks identified at the start of each phase and during milestone reviews.	Sibin Jacob
Qualitative Risk Analysis	Risks prioritized using the probability and impact matrix (Vicente, 2024). Each identified risk will be rated for severity and likelihood to determine its exposure. For clarity, risks wills be categorized as technical, financial, external, etc.	Elizabeth
Quantitative Risk Analysis	A method of risk assessment that uses numerical data and statistical techniques to quantify likelihood and impact of potential risks (Project Management Academy, 2025). Monte Carlo and severity analysis will be performed for high-priority risks. Schedule and costs are accounted for with these tools.	Chamroeun
Risk Response Technique	Strategies developed for both threats and opportunities. These choices include avoidance, mitigation, transfer, and acceptance for threats, enhancement, exploitation, and sharing opportunities (Nizhebetskyi, 2023).	Sibin Jacob
Risk Monitoring and Control	Continuous risk tracking through regular status meetings, phase reviews, and risk register updates (PMI, 2025). New risks are added. Indicators and triggers are monitored.	Raji

ROLES AND RESPONSIBILITIES

Risk Management Activity Role	Risk Management Responsibility	Assignment	Risk Attitude
Risk Manager	Oversees the entire risk process.	Elizabeth	Low
PMO (Project Management Officer)	Supports consistent risk management practices to align with organization risk policies and assists in documentation (Project Management Academy, 2025).	Alu	Low
Cost Controller	Defines and tracks the project's costs. Provides input for quantitative risk analysis.	Chamroeun	Low
Scheduler	Scheduling project timeline and identifying risks	Raji	Low
Stakeholder Engagement Lead	Stakeholder communication about risks. Supports risk manager in compliance and regularly contributes to risk reporting,	Tom	Low
The Project Manager	Responsible for planning, executing, and overseeing projects to ensure they are completed on time.	Sibin	Low
Risk Analyst	Professional who evaluates and manages potential risks that can impact an organization's finances, operations, or reputation (Project Management Academy, 2025)	Lucy	Low

BUDGETING

Using the qualitative probability and magnitude from the Risk Register, the Expected Monetary Value for each threat-type risk and total is estimated for our contingency reserve, which covers known-unknowns tied to the risks identified analyzed in the risk register. The amount is tracked and requested through a formal change request process, which requires approval by the sponsor and project manager (PMI, 2025).

EMV (Expected Monetary Value) = Probability x Impact

Table below explaining EMV-based estimate:

Risk Description	Probability	Impact ($)	EMV
Integration compatibility with Jupiter Systems	Medium (0.5)	$8000	$4000
Staff Turnover of LSDS or ASDG	Medium (0.4)	$6000	$2400
Inadequate access to LSDS data	High (0.8)	$7000	$5600
Jupiter's misalignment with LSDS	Medium (0.5)	$5000	$2500
Failure to deliver training	Low (0.2)	$10000	$2000
Negative Feedback from LSDS	Medium (0.4)	$6000	$2400
LSDS Project Managers not accepting ASDG training	Medium (0.4)	$5000	$2000
Total Contingency Reserves			$20,900

<u>Reserve Usage Protocol:</u>

Contingency Reserve: When a documented risk from the risk register requires approval by the project manager with documentation and justification.

Management Reserves: Only if unforeseen risks or strategic decisions are outside of the scope of identified risks.

TIMING

Risk Description	Probability	Time Impact (Days)	Contingency Time (Days)
LSDS's Project Managers not accepting ASDG training	Medium (0.4)	5 days	2 days
Staff Turnover	Medium (0.4)	4 days	1.6 days
Access Delays To LSDS Data	High (0.8)	7 days	5.6 days
Misalignment with Jupiter's values	Medium (0.4)	3 days	1.2 days
Failure to Deliver Training	Low (0.2)	7 days	1.4 days
Negative Feedback from LSDS	Medium (0.4)	4 days	1.6 days
Compatibility with Integration	Medium (0.5)	6 days	3.0 days
Total Schedule Contingency			16.4 days

Contingency Time (Days) = Probability x Time Impact (Days)

<u>Reserve Usage Protocol</u>

Schedule Contingency: Proactively applied after a known risk occurs. The Project Manager (PM) needs to adjust the schedule and present a justification to the project sponsor.

Schedule Management Reserve: Requires selection from the sponsor and Project Manager (Project Management Academy, 2025). SMR only approves critical impacts to scope, client-driven changes, or emerging risks.

RISK CATEGORIES

Technical Risks: Encompasses technological failures, cyberattacks, data breaches, or the obsolescence of technology (MetricStream, 2025).

- Issues with integration between Jupiter Systems and LSDS.
- Compatibility issues with current infrastructure
- Implementation of new tools

Financial Risks: Risks related to financial market, investment, and current exchange rates (MetricStream, 2025).

- Unexpected costs with training
- Overbudget
- Budget adjustments due to tools or delays

Strategic Risks: Relates to the organization’s overall goals and objectives, such as from failed business decisions, market changes, or changes in the competitive landscape (MetricStream, 2025).

- Training program fails to meet LSDS’s expectations
- ASDG’s performance affects their reputation and future opportunities

Organizational Risks: Internal structure and functions of the organization, which includes lack of stakeholder engagement alignment, internal resistance to change, and miscommunication between departments (MetricStream, 2025).

- Poor coordination between ASDG and LSDS departments
- LSDS project managers

Reputational Risks: affect public’s perception of either company and impact relationships with partners, clients, and stakeholders.

- Negative feedback from public or LSDS team about ASDG’s performance
- Failure to deliver an adequate or robust training program.

Compliance Risks: Failure to adhere to regulatory requirements and industry standards (MetricStream, 2025).

- Mishandling of classified information from LSDS
- Documentation of risk procedures are incomplete or fail to meet LSDS’s audit requirements.

Operational Risks: Potential for financial loss or disruption in business operations due to external events, human errors, or internal failures (MetricStream, 2025)

- Staff turnover for either ASDG or LSDS teams
- Lack of access to system or workplace needed for training
- Security breaches
- Scheduling issues

Opportunities:

- Increase collab with future contracts
- Development of reusable risk training

DEFINITIONS OF RISK PROBABILITY AND IMPACT

Defined Conditions for Impact Scales of a Risk on Major Project Objectives
(Examples are shown for negative impacts only)

Project Objective	Relative or numerical scales are shown				
	Very low /0.05	Low /0.10	Moderate /0.20	High /0.40	Very high /0.80
Cost	Insignificant cost increase	< 10% cost increase	10 – 20% cost increase	20 – 40% cost increase	> 40% cost increase
Time	Insignificant time increase	< 5% time increase	5 – 10% time increase	10 – 20% time increase	> 20% time increase
Scope	Scope decrease barely noticeable	Minor areas of scope affected	Major areas of scope affected	Scope reduction unacceptable to sponsor	Project end item is effectively useless
Quality	Quality degradation barely noticeable	Only very demanding applications are affected	Quality reduction requires sponsor approval	Quality reduction unacceptable to sponsor	Project end item is effectively useless

This table presents examples of risk impact definitions for four different project objectives. They should be tailored in the Risk Management Planning process to the individual project and to the organization's risk thresholds. Impact definitions can be developed for opportunities in a similar way.

(PMBOK, p. 318)

PROBABILITY AND IMPACT MATRIX

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(PMBOK, p. 331)

STAKEHOLDERS RISK TOLERENCES

STAKEHOLDER REGISTER					
Name/Group	Project Role	Risk Attitude	Major Requirements	Major Expectations	Potential Influence
Maxwell Maximillian LSDS CIO	Customer Project Sponsor	Low – Expectations are on results and strategy. Prefers minimized risks. Aligns with reliable delivery and reputation.	Personnel project risk management training and support.	Expects training will add value to personnel by equipping them to manage uncertainties in this and other projects.	If pleased with ASDG work, could influence obtaining future contracts. Could influence ASDG reputation among other corporations.
Bill Baumbas LSDS most senior project manager	LSDS primary contact	High – accepts and manages risks. Shows an openness to change and is willing to engage is deeper level training.	Training materials, documented procedures, and templates	Expects he and his colleagues will learn considerably more about project risk management.	Could have direct influence on Maxwell's perception of ASDG performance. Could have direct influence on colleague's perspectives.
Harry, Karrie, Fred, Fran, Larry, Lucy, Bart, and Barry LSDS project managers	LSDS project managers who will training and working with directly	High – has a growth mindset, the willingness the learn and apply risk management practices. They will implement changes.	Training materials, documented procedures, and templates, hands-on use and guidance of use of these	Expect they will learn considerably more about project risk management and have an opportunity to practice what is trained.	Arguably the most influential since it will be their evaluations of ASDG performance that get forwarded first to Bill and then Maxwell.
Tom Terrifick	Your boss	Medium – balance of	Excellent performance reports from LSDS CIO.	Expects ASDG excellent performance to garner future LSDS contracts	Could influence your next promotion and future advancement. One of you

ASDG Program Manager (a Senior VP)		caution and calculations.		and further ASDB reputation in the industry	key sources of advice and guidance
Your team	Tom and Raji will be working on site with you. Elizabeth, Alu, and Chamroeun will be backup resources in the office to support research and materials development	Low - My team is directly focused on support and execution of the project. We require transparent instructions, easy access to data, and a structured work environment.	Tom and Raji need onsite work area where they can to their work individually and with the client. Access to a classroom they can schedule. Access to LSDS sensitive information. Elizabeth, Alu, and Chamroeun need clear direction from the field to produce their deliverables	Expect to be treated professionally and given latitude to make independent decisions for up to the moderately critical issues. Expect clear channels of communication with LSDS as well as ASDG. Elizabeth, Alu, and Chamroeun also expect clear communication channels to obtain clarification of deliverable requirements	Directly influence ASDG performance on this contract. Directly influence relationships with all LSDS personnel. They are the ASDG face LSDS sees.
Jim Jagger Jupiter Systems COO	LSDS liaison with Jupiter Systems; owns the operating system developers	High – in coordination and innovation.	Requires close coordination with LSDS Key Stakeholders	Expects LSDS to effectively manage the project	Can influence the LSDS performance evaluations of ASDG; Could influence ASDG reputation among other corporations.
Ada Cody	Director of Jupiter Systems Software Development; manages the operating system developers	Low - development execution and ensuring that Jupiter Systems is aligned with LSDS.	Requires close coordination with LSDS Key Stakeholders	Expects LSDS to effectively manage the project	Can influence the LSDS performance evaluations of ASDG
William Whizzer	Senior Software Developer; Jupiter	Low – prioritizes clarity	Requires close coordination with	Expects LSDS to effectively manage the	Serves as conduit of information to Jim and

	Systems Software Technical Lead and onsite Jupiter Representative	and predictability over taking risks. Has a cautious approach	LSDS Key Stakeholders and daily participation with the project team	project; expects to be heard	Ada, thus can influence those who influence the LSDS performance evaluations of ASDG

RISK REGISTER

1	2	3	4	5	6	7	8	9	10	11
Date Risk Identified	**Description of Risk**	**Risk Category**	**Constraint Impacted**	**T / O**	**Probability of Occurring**	**Magnitude of Effect**	**EMV**	**Response Strategy and Plan**	**Owner**	**Status**
4/20	LSDS project managers not accepting new approaches introduced by ASDG risk training.	Organizational	Quality, schedule	T	Medium	High	High	**Mitigate** - Conduct early engagement meetings, provide adaptation support materials.	ASDG Risk Training Lead	In Progress
4/20	Compatibility with integration of Jupiter Systems and LSDS.	Technical	Cost, time	T	High	High	High	**Mitigate** - Schedule technical workshops.	ASDG Technical Lead	Monitorin g
4/20	High engagement with project managers resulting in templates reused for future contracts.	Strategic	Quality, cost	O	Medium	Medium	Medium	**Enhance** - Use standardized templates and gather feedback for improvements.	ASDG PMO	Open
4/20	Potential staff turnover for either or both ASDG and LSDS teams	Operational	Time, quality	T	Medium	High	Medium	**Mitigate -** Cross-train key staff members	HR Leads	Monitorin g

4/20	Inadequate access to LSDS's sensitive information/data, causing delay for training content development.	Operational	time	T	High	Medium	High	**Mitigate -** Define secure access protocols early.	Data Manager s	In Progress
4/20	Stakeholder's both positive and improvement feedback leads to a strong performance evaluation for ASDG teams.	Strategic	Reputation	O	Medium	High	High	**Enhance-** Showcase lessons learned from past projects and in future proposals. Create feedback loops.	ASDG Team Lead	Open
4/20	Staff of Jupiter Systems misalignment with LSDS's expectations, delivery miscommunication.	Organizational	Quality	T	Medium	Medium	Medium	**Mitigate -** Clarify roles and expectations through meetings.	Project Manager	In Progress
4/20	Newly developed risk training modules for other internal projects by LSDS	Strategic	Time, cost	O	Medium	Medium	Medium	**Exploit -** Provide documentation and training framework for reusage.	LSDS Training Coordina tor	Open
4/20	Failure to deliver an adequate training program	Reputational	Quality, reputation	T	Low	High	High	**Mitigate -** Gather feedback and adjust curriculum as needed.	ASDG Training Lead	Monitorin g

4/20	LSDS's negative feedback about ASDG's performance	Reputational	Quality, reputation	T	Medium	High	High	**Mitigate -** Set clear evaluation metrics early and continue to maintain ongoing communication with LSDS team.	ASDG Manager	In Progress
5/10/25	Cross collaboration between LSDS, ASDG, and Jupiter Systems leads to best practices and innovation adopted into future and current projects.	Strategic	Quality, time	O	Medium	High	High	**Enhance** - Document and share success stories, encourage regular knowledge joint perspective sharing sessions.	ASDG Project Manager	In Progress

REPORTING FORMATS

The primary format used to communicate risks for this project is the Risk Register, which includes the following details for each entry:

- Date identified
- Description
- Category
- Constraints
- Type (Opportunity or Threat)
- Probability
- Magnitude of Effect
- Expected Monetary Value (EMV)
- Response Strategy & Plan
- Owner
- Risk Status

The risk register is in a structured table format and is regularly updated. High-risk items are visibly highlighted. Updates are tracked by the “Status” column. The outcomes of risk management processes are emerging risks, mitigation actions taken, probability and impact, and lessons learned.

They are, documented in the risk register and project documentation, analyzed during project review meetings with key stakeholders to address trends and evaluate effectiveness, communicated through biweekly status updated and shared with LSDS contacts and ASDG leaders, and are part of the final documentation. Lessons learned are summarized in the Project Closure Report, with emphasis that includes successful mitigation strategies, challenges faced, and recommendations for future projects (PMI, 2021).

TRACKING

All risk-related activities are documented and tracked through the risk register, which is the central repository for all identified risks, analysis, response strategies, and updates throughout the project timeline (PMI, 2021).

The Risk Register will be updated on a regular basis by the assigned risk responsibilities and reviewed during project meetings. Updates will include response actions taken, status changes, or any newer risks. For traceability and transparency, each risk will have information such as probability, impact, mitigation progress, or ownership changes.

Risk management processes will be evaluated by:

- Risks were identified and assessment in a timely manner
- Response plans and mitigation executed as intended
- New risks documented appropriately
- Risk outcomes affected by stakeholders

Missed or mishandled managed risks will be flagged during reflections or post-project reviews, as this will examine if the correct management tools and techniques were implemented and to identify gaps in the plan ().

Findings from the audit will be used to guide future risk management practices in LSDS and ASDG. They will be recorded in the project's Lessons Learned document for reference.

QUALITY IMPROVEMENT METHODOLOGIES

Quality improvement methodologies are essential for identifying and managing project risks, ensuring deliverables meet stakeholder expectations and prevent defects (AtlexSoft, 2023). With risk management integration, there are two types of Quality Improvement Methodologies to be used: Quality Assurance and Quality Control.

Quality Assurance (QA) focuses on process-oriented, proactive measures to prevent failures or defects in project deliverables (AtlexSoft, 2023). QA implements risk-aware processes, which ensures they are built with risk controls, safeguards, and checkpoints. It serves as an early identification of process-based risks through planning, process audits, and standard adherence checks. QA helps surface risks related to poor documentation, communication gaps, or workflow inefficiencies. In addition, Quality Assurance encourages key principles such as feedback loop and retrospectives, which supports lessons learned. For example, during training development for LSDS. the QA methods such checklists. peer reviews, and review gates ensure the quality and alignment of deliverables.

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Quality Control (QC) is reactive and product-oriented, as it focuses on detecting and correcting defect. QC verifies deliverables against requirements by identifying errors before the deliverables reach stakeholders, reducing delivery and reputational risks. Through frequent inspections and testing cycles, QC detect emerging risks, which can be documented in the risk register. When defects are detected, QC triggers corrective actions such as reallocation of resources, rework, and/or timeline adjustments. If a training module fails to meet LSDS standards during a run through, QC will prompt the team to identify the root causes and take immediate action to manage the risk before deployment.

Both Quality Assurance and Quality Control create a reiterative loop that reduces uncertainty, enhance stakeholder confidence, and supports the risk management objectives of maximizing opportunities and minimizing threats. Integrating both methodologies allow ASDG and LSDS to continuously arrange quality-related risks to aim for a successful delivery of the project's training and outcomes.

Project Risk Management Plan Approvals

The undersigned acknowledge they have reviewed the *<LSDS>* Project Risk Management Plan and agree with the approach it presents. Changes to this Project Risk Management Plan will be coordinated with and approved by the undersigned or their designated representatives.

Signature:	Sibin Jacob	Date:	6/2/25
Print Name:	Sibin Jacob		
Title:	Project Manager		
Role:	Lead Project Manager ASDG		

Signature:	Tom Terrifick	Date:	6/2/25
Print Name:	Tom Terrifick		
Title:	ASDG Program Manager (a Senior VP)		
Role:	Head of ASDG		

Signature:	Maxwell Maximillian	Date:	6/2/25
Print Name:	Maxwell Maximillian		
Title:	LSDS CIO		
Role:	Project Sponsor, LSDS		

APPENDIX A: REFERENCES

Document Name and Version	Description	Location
PMBOK, 5th ed.	Guide to the Project Management Body of Knowledge	Project manager's personal desk copy.
Cflow (2025, March 24)	Project Risk Management: A Comprehensive Guide For Project Managers	https://www.cflowapps.com/project-risk-management-guide/
MetricStream. (2025).	*What are risk categories? Types and ways to determine them.*	https://www.metricstream.com/learn/what-are-risk-categories.htmlMetricstream
Nizhebetskyi, D. (2023, September 17).	*Risk response strategies (Definitive guide with examples).* IT PM School.	https://itpmschool.com/risk-response-strategy/
Stanford University IT. (2025).	*Risk Classifications*	https://uit.stanford.edu/guide/riskclassifications
Project Management Academy. (2025).	*Risk Management Process For PMT*	https://projectmanagementacademy.net/resources/risk-management-process-for-pmp/
Vicente, V. (2024, February 15).	*Risk Assessment Matrix: Overview and Guide*	https://www.auditboard.com/blog/what-is-a-risk-assessment-matrix/
Aldridge, E. (2022). Project Management Academy	*Using Contingency and Schedule Reserve in Your Projects.*	*https://projectmanagementacademy.net/resources/blog/using-contingency-and-schedule-reserve-in-your-projects/*
AltexSoft. (2023).	Quality Assurance (QA), Quality Control and Testing.	https://www.altexsoft.com/whitepapers/quality-assurance-quality-control-and-testing-the-basics-of-software-quality-management/

APPENDIX B: KEY TERMS

Term	Definition
Risk	Used synonymously with the term uncertainty
Uncertainty	Used synonymously with the term risk
Risk Attitude	A person’s or organization’s chosen response to uncertainty that can impact project objectives (PMI, 2025)
EMV (Expected Monetary Value)	Calculated by probability * impact. The calculation helps determine the average financial gain or loss expected from a decision to consider different outcomes.
Probability	Likelihood of a specific outcome occurring
Impact	The extend of an event affecting a risk organization